

DATA ALCOTT SYSTEMS

Free Web Development Internship Online Task Report

Task ID: WD-CRM-005

Project Name	Marketing Automation System	Domain	CRM - Marketing Automation
Company	Data Alcott Systems	Internship Type	Free Online Internship
Architecture	Client-Side (No Database)	Color Scheme	Red (#DC2626), White, Gray
Primary Technologies	HTML5, CSS3, JavaScript	Submission Date	October 2026

1. Executive Summary & Objectives

The **Marketing Automation System (CRM)** is a frontend-centric web application engineered to empower marketing teams to manage, execute, and analyze automated campaigns without reliance on backend database servers. By leveraging pure JavaScript array structures, object references, and local persistent storage, the system delivers high responsiveness, fluid interactivity, and a clean, marketing-oriented user experience.

2. System Modules & Functional Breakdown

In accordance with the project guidelines, the application implements all mandatory operational modules:

- **Dashboard:** Displays key metric cards (Total Campaigns, Active Subscribers, Average Open/Click Rates) alongside real-time visual charts for engagement performance and audience growth dynamics.
- **Campaign Management (CRUD):** Full lifecycle management allowing creation, editing, status updates (Draft, Scheduled, In Progress, Completed, Cancelled), and deletion with live search and filtering.
- **Email Template Studio:** Template catalog with a live modal preview featuring desktop/mobile switching and simulated test email dispatches.
- **Subscriber List & Segmentation:** Segment-based audience records (VIP, Engaged, Leads, Trial) with contact record additions and CSV export capabilities.

- **Campaign Analytics & ROI Calculator:** Complete breakdown of delivery rates, open/click metrics, device breakdown, and an interactive conversion revenue simulator.
- **Automation Workflows:** Step-by-step visual dripline sequence designer configured for customer onboarding and cart recovery funnels.
- **Landing Page Designer:** Interactive WYSIWYG editor with live browser-canvas preview and instant color theme alterations.
- **Reports & Audit Log:** Systematic campaign audit logs with PDF generation/printing and CSV data exports.
- **Simulated User Authentication:** Profile management suite maintaining active session roles and profile credentials.

3. Bonus Features & UI/UX Highlights

Feature	Implementation Details	Status
Data Visualization	Integrated Chart.js line, bar, and doughnut charts for real-time trend analytics.	Completed
Simulated A/B Testing	Variant comparison tool calculating projected conversion gains between subject lines.	Completed
Dark Mode Toggle	High-contrast, eye-friendly theme with persistent local storage retention.	Completed
Email Preview & Testing	Responsive iframe-style preview with simulated test delivery alerts.	Completed
Export Utility	One-click CSV subscriber exports and print-ready PDF reporting layouts.	Completed
Data Persistence	Synchronized <code>localStorage</code> caching for campaigns, subscribers, and profile settings.	Completed

4. 7-Day Development Lifecycle Summary

1. **Day 1 (Setup & Planning):** UI wireframing, color palette definition (#DC2626), and data structure design.
2. **Day 2 (HTML Structure):** Semantic layout creation, sidebar navigation, modular panels, and modal scaffolds.
3. **Day 3 (CSS Styling):** Desktop layout development, custom design system, card styling, and hover animations.
4. **Day 4 (Responsive Design):** Media query integration, mobile sidebar sliding menu, and cross-device testing.

5. **Day 5 (JavaScript Development):** Implementation of CRUD algorithms, search/filter logic, and state controllers.
6. **Day 6 (Testing & Enhancements):** Integration of Chart.js, A/B testing suite, ROI simulator, and dark theme support.
7. **Day 7 (Deployment & Submission):** Documentation compilation, code commenting, and preparation for deployment on GitHub Pages/Vercel.

5. Submission Deliverables & Links

- **Task Link on Blog:** <https://www.freeinternships.in/blog/>
- **GitHub Repository:** <https://github.com/omchoudharydev/Marketing-Automation-System-CRM-.git>
- **YouTube Demo Video:** https://youtu.be/cmriPrUmEIs?si=ciXblvHi0JFt_wQp
- **Live Application Link:** <https://marketingautomationsystemcrm.netlify.app>

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